# Skill Builder Pack 3 - Blending Data in Excel

## Purpose

Once data has been imported and cleaned, the next step in the analysis process is often **data blending**.

Data blending means **combining information from multiple datasets so that they can be analysed together**.

In many organisations, relevant information is stored in separate systems. For example:

- Attendance data may come from a learning platform
- Student information may come from a student records system
- Engagement data may come from a reporting dashboard

To gain useful insights, these datasets must be **combined into a single view**.

In Excel this can be done in two common ways:

- Using **Power Query**
- Using **XLOOKUP**

Both approaches are useful, and understanding when to use each method is an important skill for data analysts.

## Learning Outcomes

By the end of this skill builder you should be able to:

- Understand the concept of **data blending**
- Combine datasets using **Power Query**
- Combine datasets using **XLOOKUP**
- Understand the advantages of each method

# Starting Point

You should now have an Excel workbook containing the cleaned datasets from previous skill builders:

- Attendance data
- Student details
- Engagement scores
- Withdrawal records

Each dataset currently exists as a **separate table**.

Your goal is to **combine these datasets so that information about each student can be viewed together**.

## Key Field

In order to combine datasets, a shared identifier is required.

In this dataset, the shared field is:

**student_id**

This value appears in every dataset and allows records to be matched correctly.

# Method 1 – Blending Data Using Power Query

Power Query allows datasets to be combined automatically using a process called **Merge**.

This approach is commonly used when working with **larger datasets or automated data pipelines**.

## Example

Attendance data contains:

| student_id | module | attendance_status |
|---|---|---|
| 1001 | Data Fundamentals | Present |

Student details contain:

| student_id | first_name | last_name |
|---|---|---|
| 1001 | Ali | Khan |

By merging the datasets using **student_id**, the combined dataset becomes:

| student_id | first_name | last_name | module | attendance_status |
|---|---|---|---|---|

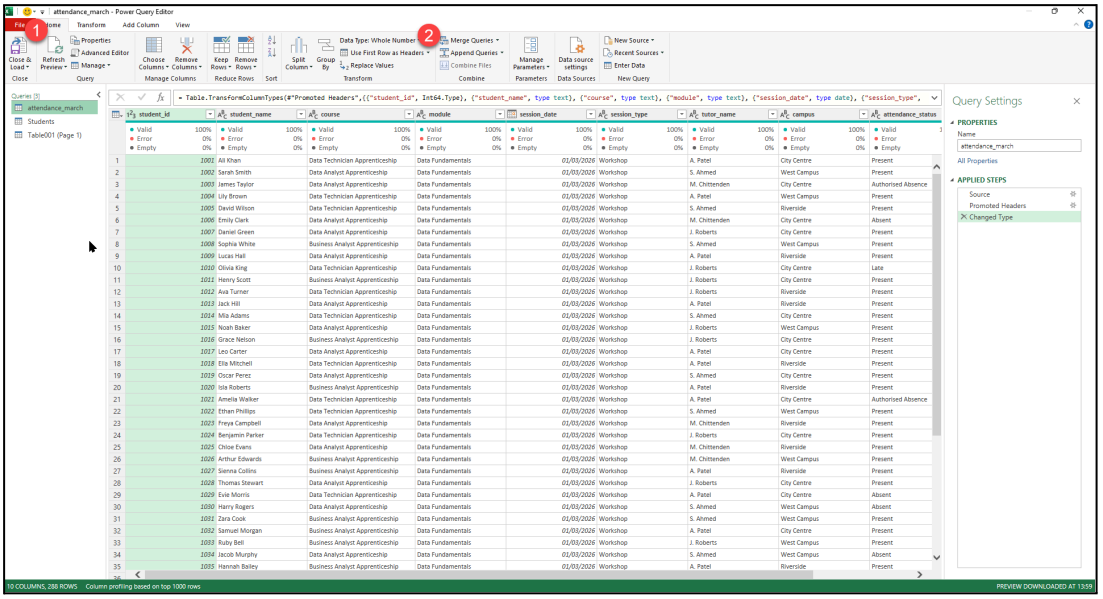
# Practice Activity

Open the **attendance dataset** in Power Query by either double clicking on one of the queries in the **Data > Queries & Connections** option, or select **Data > Get Data > Launch Power Query Editor** .

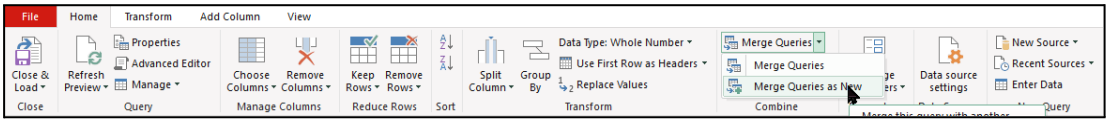
Follow these steps:

1. Select **Home**

2. Choose **Merge Queries**




3. Select **Merge Queries as New**

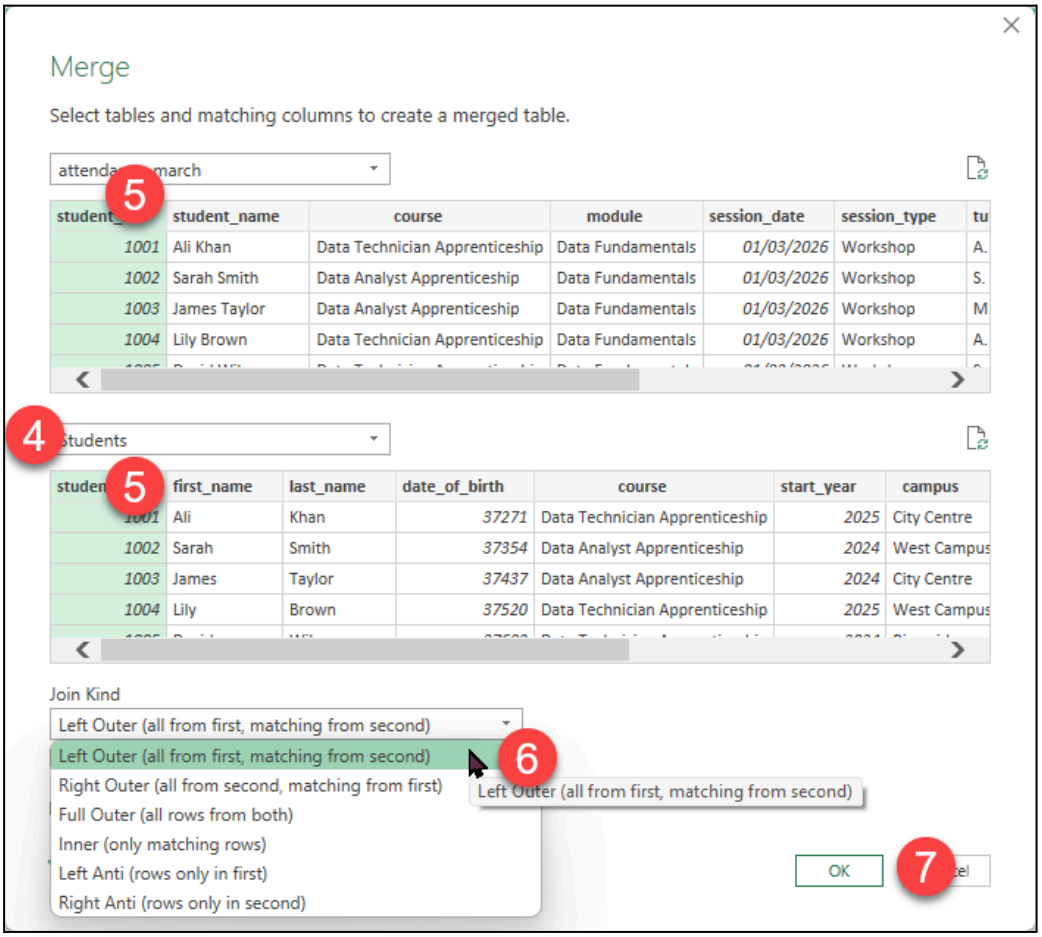



4. Make sure that **attendance_march** is selected in the top box, and then select the **student details table** in the lower box. These will be the data sources we will merge

5. Choose **student_id** as the matching column

6. Select **Left Join.** Notice how the system automatically calculates the matches available

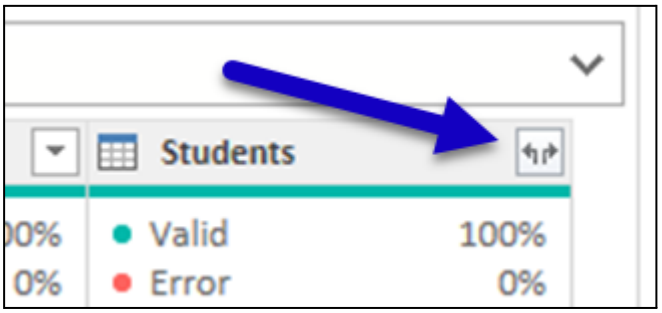
7. Click **OK**



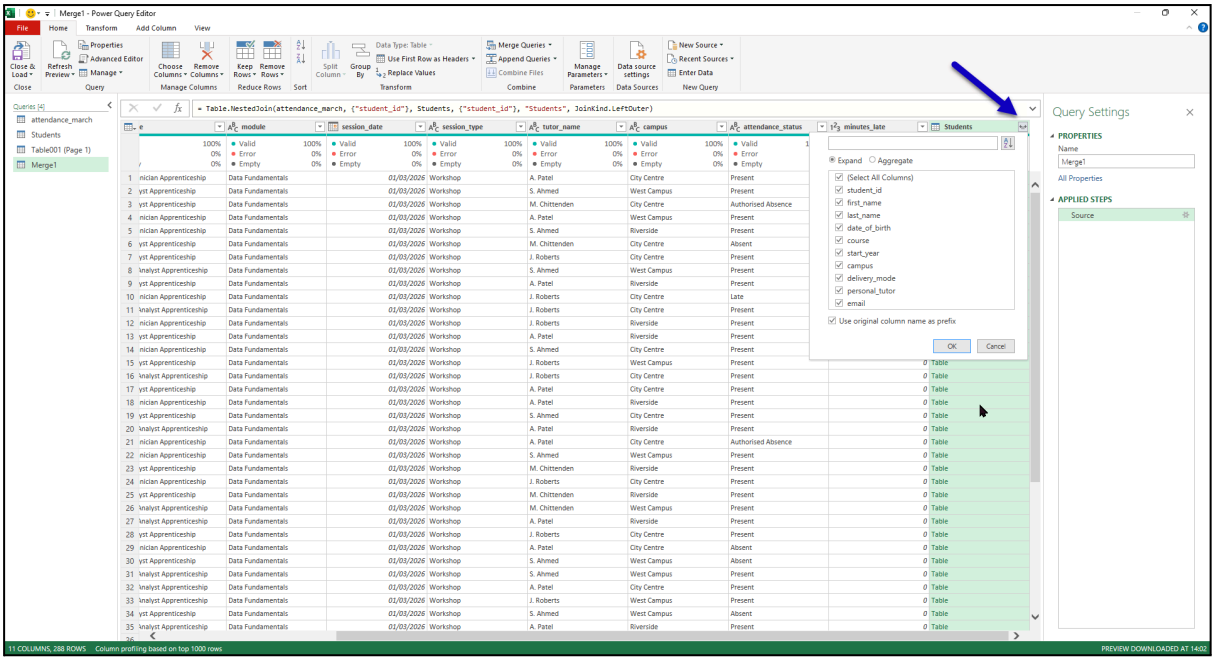

Once merged:

- Scrub over to the far right of the table where the new column has been added - currently it should just say *Table*. This is because Power Query knows you want to merge that table, but it does not know what specific rows you want

- Expand the columns from the student details dataset by clicking on the icon that looks like two arrows pointing in opposite directions




- When this expands, choose everything except **student_id, course, campus** because these already exist in the table so we would be duplicating information

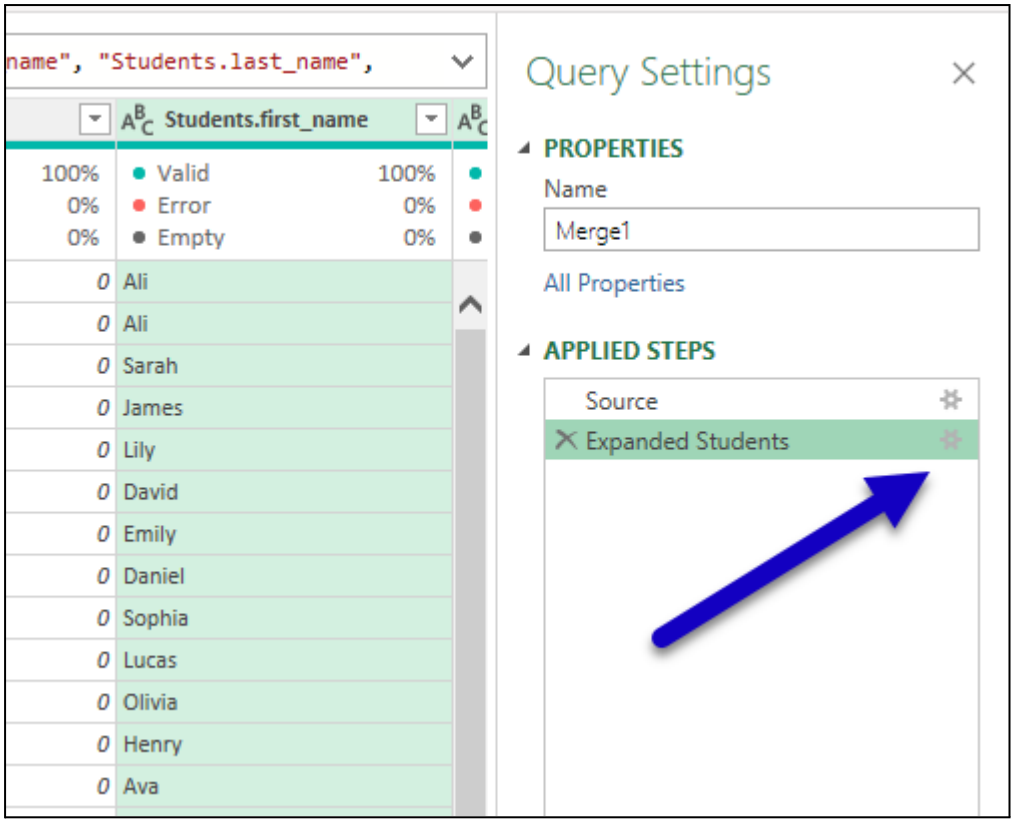



You can choose to *Use original column names as prefix* as you see fit, if you want to sudit where the data came from

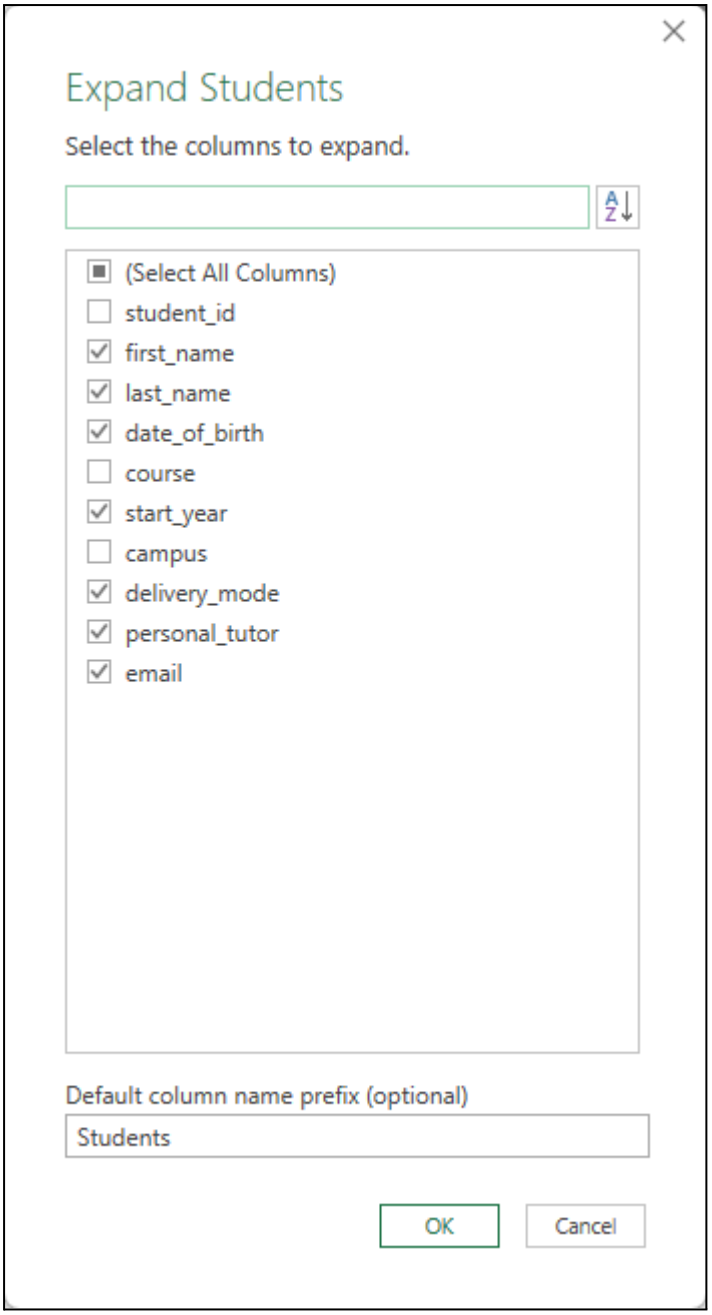
Click **OK**

You have now combined two datasets.

Note - this step will now appear in the **Query Settings** panel on the right with a small gear cog next to it.




This gear cog will allow you to go back in and select different columns if you want - click on it and try it. You will see this screen.




## Adding Engagement Data

Repeat the merge process to add the **engagement scores dataset, picking the fields that can be used to avoid duplication**.

The final dataset should now contain:

- Student name
- Module
- Attendance status
- Engagement score

Once you have done this click on **Close and Apply**

# Method 2 – Blending Data Using XLOOKUP

Excel also allows datasets to be combined using formulas.

One of the most useful functions for this purpose is **XLOOKUP**.

XLOOKUP searches for a value in one table and returns a matching value from another table.

## Example Formula

=XLOOKUP(student_id, student_table[student_id], student_table[first_name])

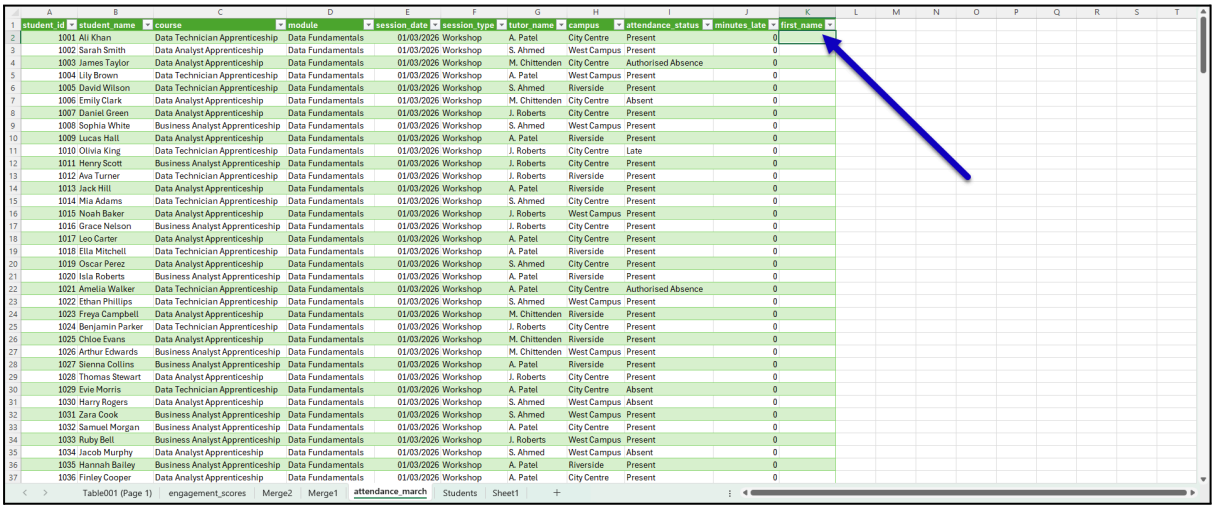
This formula:

1. Looks for the student_id
2. Finds the matching row in the student table
3. Returns the student's first name

## Practice Activity

In the attendance table:

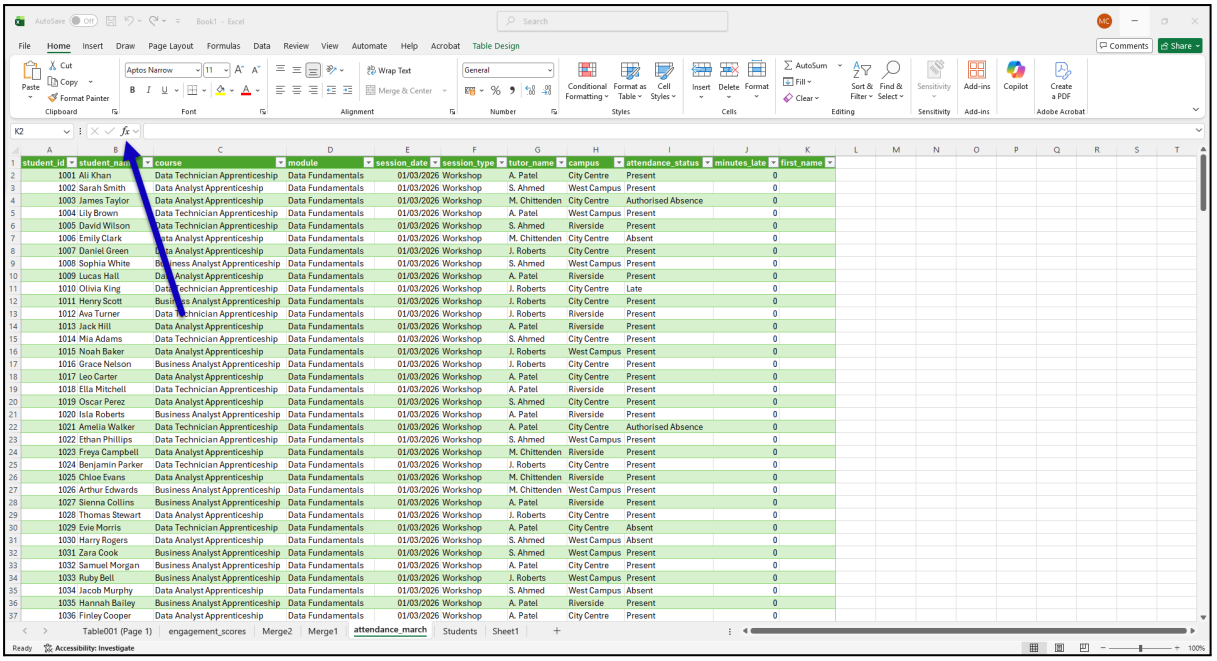
Create a new column called **first_name**.



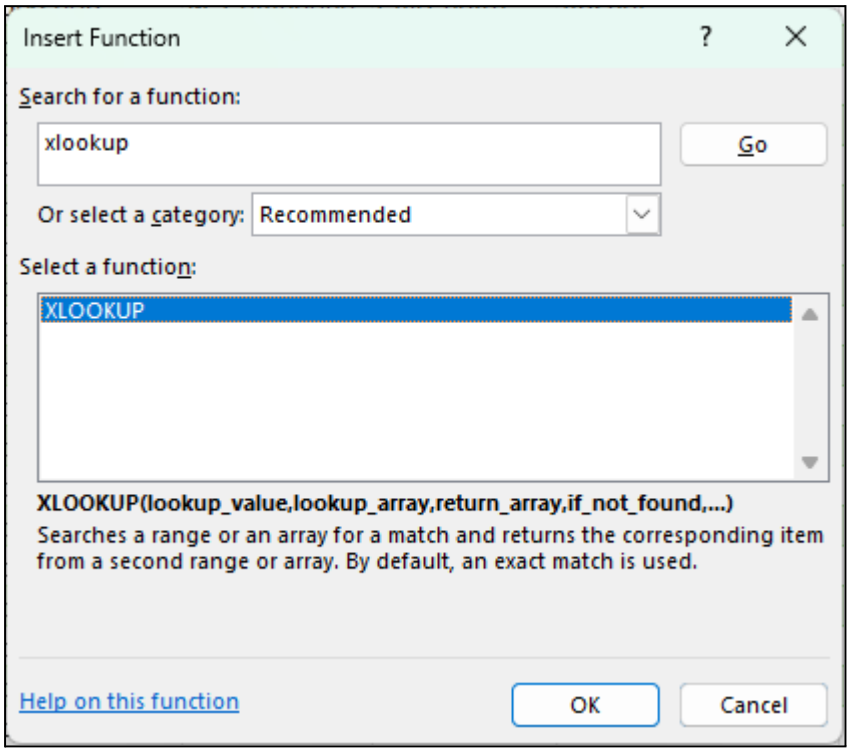
| student_id | student_name | course | module | session_date | session_type | tutor_name | campus | attendance_status | minutes_late | first_name |
|---|---|---|---|---|---|---|---|---|---|---|
| 1001 | Ali Khan | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | City Centre | Present | 0 | |
| 1002 | Sarah Smith | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | West Campus | Present | 0 | |
| 1003 | James Taylor | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | M. Chittenden | City Centre | Authorised Absence | 0 | |
| 1004 | Lily Brown | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | West Campus | Present | 0 | |
| 1005 | David Wilson | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | Riverside | Present | 0 | |
| 1006 | Emily Clark | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | M. Chittenden | City Centre | Absent | 0 | |
| 1007 | Daniel Green | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | City Centre | Present | 0 | |
| 1008 | Sophia White | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | West Campus | Present | 0 | |
| 1009 | Lucas Hall | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | Riverside | Present | 0 | |
| 1010 | Olivia King | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | City Centre | Late | 0 | |
| 1011 | Henry Scott | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | City Centre | Present | 0 | |
| 1012 | Ava Turner | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | Riverside | Present | 0 | |
| 1013 | Jack Hill | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | Riverside | Present | 0 | |
| 1014 | Mia Adams | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | City Centre | Present | 0 | |
| 1015 | Noah Baker | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | West Campus | Present | 0 | |
| 1016 | Grace Nelson | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | City Centre | Present | 0 | |
| 1017 | Leo Carter | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | City Centre | Present | 0 | |
| 1018 | Ella Mitchell | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | Riverside | Present | 0 | |
| 1019 | Oscar Perez | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | City Centre | Present | 0 | |
| 1020 | Isla Roberts | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | Riverside | Present | 0 | |
| 1021 | Amelia Walker | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | City Centre | Authorised Absence | 0 | |
| 1022 | Ethan Phillips | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | West Campus | Present | 0 | |
| 1023 | Freya Campbell | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | M. Chittenden | Riverside | Present | 0 | |
| 1024 | Benjamin Parker | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | City Centre | Present | 0 | |
| 1025 | Chloe Evans | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | M. Chittenden | Riverside | Present | 0 | |
| 1026 | Arthur Edwards | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | M. Chittenden | West Campus | Present | 0 | |
| 1027 | Sienna Collins | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | Riverside | Present | 0 | |
| 1028 | Thomas Stewart | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | City Centre | Present | 0 | |
| 1029 | Evie Morris | Data Technician Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | City Centre | Absent | 0 | |
| 1030 | Harry Rogers | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | West Campus | Absent | 0 | |
| 1031 | Zara Cook | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | West Campus | Present | 0 | |
| 1032 | Samuel Morgan | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | City Centre | Present | 0 | |
| 1033 | Ruby Bell | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | J. Roberts | West Campus | Present | 0 | |
| 1034 | Jacob Murphy | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | S. Ahmed | West Campus | Absent | 0 | |
| 1035 | Hannah Bailey | Business Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | Riverside | Present | 0 | |
| 1036 | Finley Cooper | Data Analyst Apprenticeship | Data Fundamentals | 01/03/2026 | Workshop | A. Patel | City Centre | Present | 0 | |

Table001 (Page 1) | engagement_scores | Merge2 | Merge1 | attendance_march | Students | Sheet1

Now select the first cell in the column, then click on the formula button in the Excel formula bar

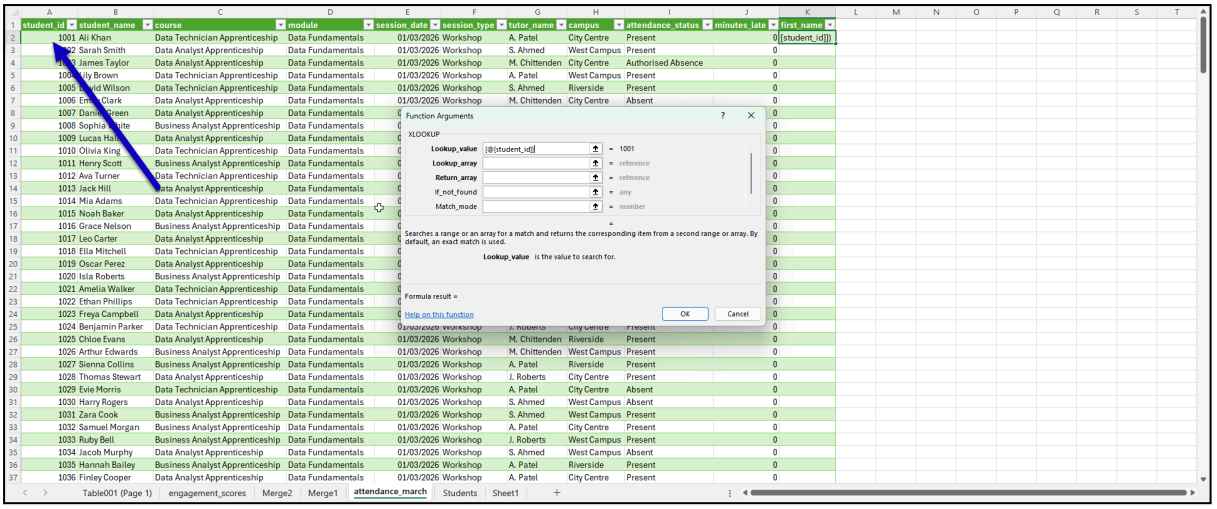


In the box that appears, type **XLOOKUP** and click on **Go**. This will now search through Excel's library to locate the **XLOOKUP** template

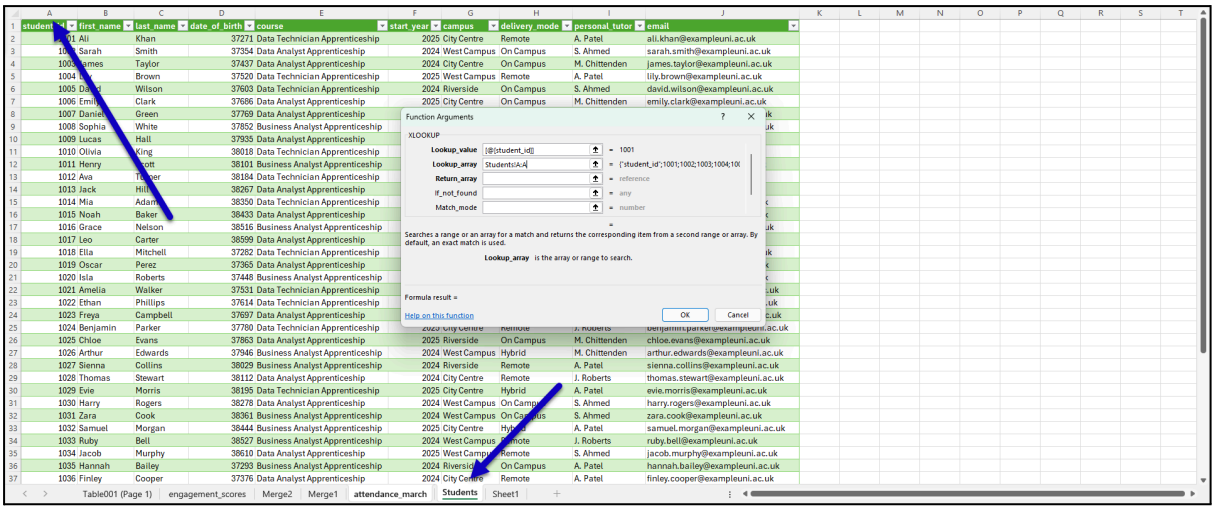



Click **OK.** This will open the Formula Wizard which you will fill in piece by pieve.

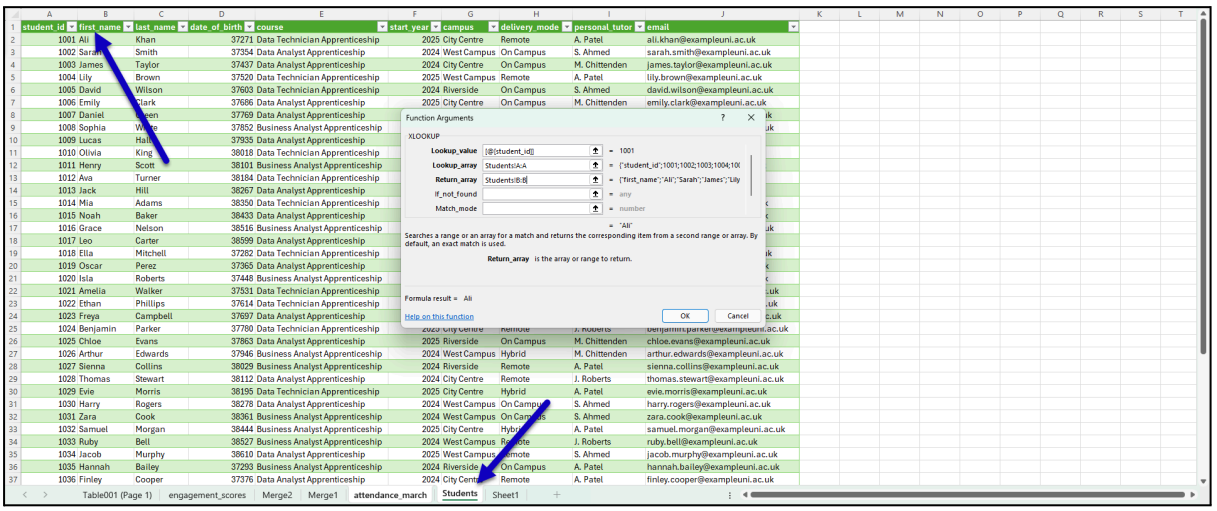
In **Lookup_value** select the field you want to match, in this case Cell **A2**



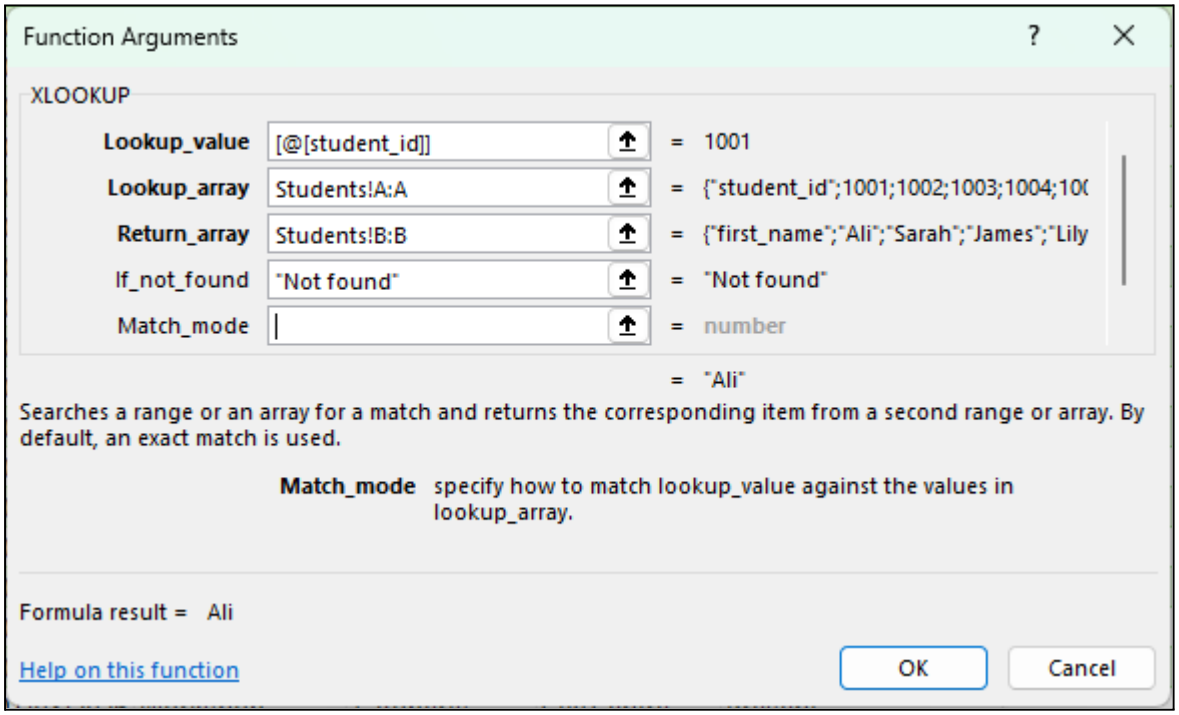
In **Lookup_array**, select the column where that match might be found, in this case the **Students** tab and Column A



In **Return_array** select the column where the information you want to retrieve in, in this case **Students > first_name**



In If_not_found (which is not in **bold** so in not mandatory) type in a message that you would put in the the information isn;t there e.g. Not found (the speech marks are added automatically).




This is how you wse **XLOOKUP** to retrieve the student's first name from the student details table.

Repeat the process to retrieve:

- last_name
- course
- engagement_score

This will add additional information to the attendance dataset.

## Comparing the Two Methods

Both methods allow datasets to be combined, but they are suited to different situations.

| Method | Best Used When |
|---|---|
| Power Query | Working with large datasets or automated workflows |
| XLOOKUP | Quick analysis within a spreadsheet |

Power Query is often preferred in professional data preparation because it allows the entire blending process to be **repeated automatically when the data is refreshed**.

XLOOKUP is useful for **quick analysis or smaller datasets** where formulas are easier to apply.

## Key Concept

Data blending allows information from different systems to be brought together so that a more complete picture can be analysed.

In this scenario, combining attendance, engagement, and student information allows the team to identify students who may be at risk of disengaging.

Without blending the data, each dataset would only provide **part of the overall picture**.